

操作系统实验报告

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实验题目：编译内核

一、实验内容

- 1、获取 Linux 6.16.12 内核源码（之前实验已获取）
- 2、基于当前系统内核配置文件，生成新内核的配置（.config），并完成自定义设置，嵌入个人标识
- 3、通过 make 命令编译内核及模块，生成可引导的内核镜像文件
- 4、安装新内核后，重启系统并验证是否成功基于 Linux 6.16.12 内核运行

二、实现步骤

1、环境检查与依赖安装

（1）更新软件源索引

```
sudo apt-get update
```

```
sudo apt-get upgrade -y
```

（2）安装核心编译工具

```
sudo apt-get install -y build-essential
```

（3）安装内核配置与依赖库

```
sudo apt-get install -y pkg-config libgtk2.0-dev libcanberra-gtk-module glade libglade2-dev
```

```
sudo apt-get install -y flex bison libssl-dev libelf-dev libncurses-dev
```

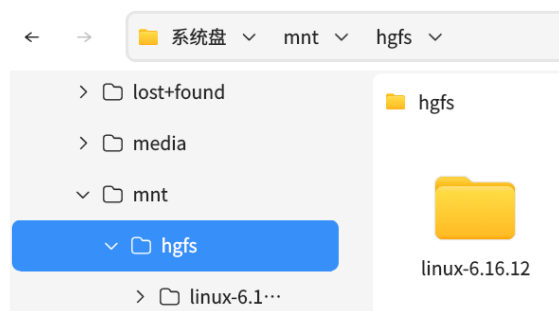
2、检查系统信息

（1）当前系统内核版本：5.10.0-15-generic

（2）检查磁盘空间：充足，编译内核使用/dev/sda6，可用 52G

```
LZH2311451@LZH2311451-pc: ~  
文件(F) 编辑(E) 视图(V) 搜索(S) 终端(T) 帮助(H)  
LZH2311451@LZH2311451-pc:~$ uname -r  
5.10.0-18-generic  
LZH2311451@LZH2311451-pc:~$ uname -a  
Linux LZH2311451-pc 5.10.0-18-generic #1~v10pro-KYLINOS SMP Mon Mar 17 11:49:02 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux  
LZH2311451@LZH2311451-pc:~$ cat /etc/os-release  
NAME="Kylin"  
VERSION="银河麒麟桌面操作系统V10 SP1"  
VERSION_US="Kylin Linux Desktop V10 SP1"  
ID=kylin  
ID_LIKE=debian  
PRETTY_NAME="Kylin V10 SP1"  
VERSION_ID="v10"  
HOME_URL="http://www.kylinos.cn/"  
SUPPORT_URL="http://www.kylinos.cn/support/technology.html"  
BUG_REPORT_URL="http://www.kylinos.cn/"  
PRIVACY_POLICY_URL="http://www.kylinos.cn"  
VERSION_CODENAME=kylin  
UBUNTU_CODENAME=kylin  
PROJECT_CODENAME=V10SP1  
KYLIN_RELEASE_ID="2503"  
LZH2311451@LZH2311451-pc:~$ df -BG  
文件系统      1G-块   已用   可用   已用%   挂载点  
udev           2G      0G      2G      0%   /dev  
tmpfs          1G      1G      1G      1%   /run  
/dev/sda5      58G     13G     42G     24%   /  
/dev/sda1       2G      1G      2G      7%   /boot  
/dev/sda6      55G     1G     52G      1%   /data  
tmpfs          2G      1G      2G      1%   /dev/shm  
tmpfs          1G      1G      1G      1%   /run/lock  
tmpfs          2G      0G      2G      0%   /sys/fs/cgroup  
tmpfs          1G      1G      1G      1%   /run/user/1000  
/dev/sr0        5G      5G      0G     100%   /media/LZH2311451/Kylin-Desktop-V10-SP1  
vmhgfs-fuse    752G    592G    161G     79%   /mnt/hgfs  
LZH2311451@LZH2311451-pc:~$
```

3、内核配置



(1) 复制当前系统的配置文件

创建软链接：将源码目录链接到/usr/src/linux

进入内核源码目录：cd /usr/src/linux

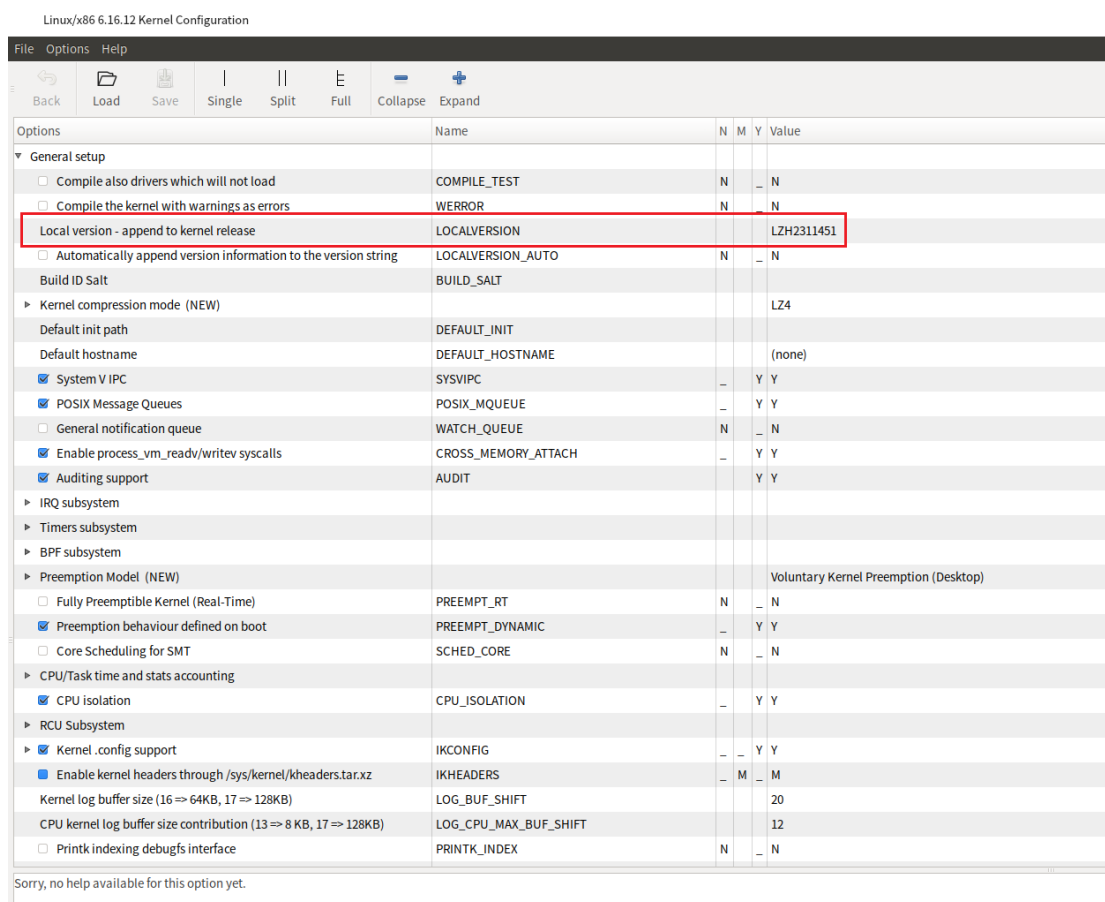
复制当前配置文件为新内核的.config：cp /boot/config-`uname -r` .config

(2) 升级配置文件：yes "" | make oldconfig

(3) 自定义配置

进入图形化配置：make gconfig

设置Local version - append to kernel release 为LZH2311451



4、编译内核

```
LZH2311451@LZH2311451-pc: /usr/src/linux
文件(F) 编辑(E) 视图(V) 搜索(S) 终端(T) 帮助(H)

LD [M] net/vmw_vsock/vsock.ko
CC [M] net/vmw_vsock/vsock_diag.mod.o
LD [M] net/vmw_vsock/vsock_diag.ko
CC [M] net/vmw_vsock/vmw_vsock_vmci_transport.mod.o
LD [M] net/vmw_vsock/vmw_vsock_vmci_transport.ko
CC [M] net/vmw_vsock/vmw_vsock_virtio_transport.mod.o
LD [M] net/vmw_vsock/vmw_vsock_virtio_transport.ko
CC [M] net/vmw_vsock/vmw_vsock_virtio_transport_common.mod.o
LD [M] net/vmw_vsock/vmw_vsock_virtio_transport_common.ko
CC [M] net/vmw_vsock/hv_sock.mod.o
LD [M] net/vmw_vsock/hv_sock.ko
CC [M] net/vmw_vsock/vsock_loopback.mod.o
LD [M] net/vmw_vsock/vsock_loopback.ko
CC [M] net/nsh/nsh.mod.o
LD [M] net/nsh/nsh.ko
CC [M] net/hsr/hsr.mod.o
LD [M] net/hsr/hsr.ko
LZH2311451@LZH2311451-pc: /usr/src/linux$ sudo make -j4
输入密码
DESCEND objtool
INSTALL libsubcmd_headers
CALL scripts/checksyscalls.sh
Kernel: arch/x86/boot/bzImage is ready (#1)
LZH2311451@LZH2311451-pc: /usr/src/linux$
```

5、安装内核

安装内核模块 `sudo make modules_install`

安装内核镜像与配置 `sudo make install`

```

LZH2311451@LZH2311451-pc:/usr/src/linux$ sudo make install
INSTALL /boot
run-parts: executing /etc/kernel/postinst.d/apt-auto-removal 6.16.12LZH2311451 /boot/vmlinuz-6.16.12LZH2311451
run-parts: executing /etc/kernel/postinst.d/dkms 6.16.12LZH2311451 /boot/vmlinuz-6.16.12LZH2311451
* dkms: running auto installation service for kernel 6.16.12LZH2311451
Kernel preparation unnecessary for this kernel. Skipping...

Building module:
cleaning build area...
make -j4 KERNELRELEASE=6.16.12LZH2311451 -C /lib/modules/6.16.12LZH2311451/build M=/var/lib/dkms/evdi/1.14.0/build DKMS_BUILD=1.....(bad exit status: 2)
Error! Bad return status for module build on kernel: 6.16.12LZH2311451 (x86_64)
Consult /var/lib/dkms/evdi/1.14.0/build/make.log for more information.

Kernel preparation unnecessary for this kernel. Skipping...

Building module:
cleaning build area...(bad exit status: 2)
make -j4 KERNELRELEASE=6.16.12LZH2311451 -f Makefile.dkms all...(bad exit status: 2)
Error! Bad return status for module build on kernel: 6.16.12LZH2311451 (x86_64)
Consult /var/lib/dkms/kvdp-control/1.0.5.1/build/make.log for more information.

Kernel preparation unnecessary for this kernel. Skipping...

Building module:
cleaning build area...
make -j4 KERNELRELEASE=6.16.12LZH2311451 -C /lib/modules/6.16.12LZH2311451/build M=/var/lib/dkms/optilauncher/1.0.1.0/build.....(bad exit status: 2)
Error! Bad return status for module build on kernel: 6.16.12LZH2311451 (x86_64)
Consult /var/lib/dkms/optilauncher/1.0.1.0/build/make.log for more information.

Kernel preparation unnecessary for this kernel. Skipping...

Running the pre_build script:

Building module:
cleaning build area...
make -j4 KERNELRELEASE=6.16.12LZH2311451 -C /lib/modules/6.16.12LZH2311451/build M=/var/lib/dkms/xtables-addons/3.9/build/extensions XA_ABSTOPSRCDIR=/var
uild -name '*.ko' -exec mv -v {} /var/lib/dkms/xtables-addons/3.9/build \;.....
Error! Build of xt_comm.ko failed for: 6.16.12LZH2311451 (x86_64)
Consult the make.log in the build directory
/var/lib/dkms/xtables-addons/3.9/build/ for more information.

[ OK ]
run-parts: executing /etc/kernel/postinst.d/initramfs-tools 6.16.12LZH2311451 /boot/vmlinuz-6.16.12LZH2311451
update-initramfs: Generating /boot/initrd.img-6.16.12LZH2311451
I: The initramfs will attempt to resume from /dev/sda7
I: (UUID=a72508e3-bdec-4af9-b2ea-f80f1dfbf5b7)
I: Set the RESUME variable to override this
run-parts: executing /etc/kernel/postinst.d/zz-update-grub 6.16.12LZH2311451 /boot/vmlinuz-6.16.12LZH2311451
Sourcing file '/etc/default/grub'
Sourcing file '/etc/default/grub.d/init-select.cfg'
正在生成 grub 配置文件 ...
找到主题: /usr/share/grub/themes/UKUI/theme.txt
找到 Linux 镜像: /boot/vmlinuz-6.16.12LZH2311451
找到 initrd 镜像: /boot/initrd.img-6.16.12LZH2311451
找到 Linux 镜像: /boot/vmlinuz-5.10.0-18-generic
找到 initrd 镜像: /boot/initrd.img-5.10.0-18-generic
完成

```

6、重启并确认新内核启动

重启系统 `sudo reboot`



```

LZH2311451@LZH2311451-pc: ~
文件(F) 编辑(E) 视图(V) 搜索(S) 终端(T) 帮助(H)
LZH2311451@LZH2311451-pc:~$ uname -r
6.16.12LZH2311451
LZH2311451@LZH2311451-pc:~$ uname -a
Linux LZH2311451-pc 6.16.12LZH2311451 #2 SMP PREEMPT_DYNAMIC Fri Nov 7 08:00:02
CST 2025 x86_64 x86_64 x86_64 GNU/Linux
LZH2311451@LZH2311451-pc:~$

```

三、结论

几经周折，最终成功。